

value of the underlying, as it will be settled for it. But when there is plenty of time left, other factors take the driver's seat.

There are multiple reasons for why the price further out in the future is different, but as quantitative traders we rarely need to dig deep into these reasons. Factors in play include interest rates, cost of storage, and seasonality.

What matters is how to interpret this pattern. Here is a simple way to think about it. If the term structure is in contango, as in Figure 8.1, there is a bearish bias. Why? Because the closer a contract gets to expiry, the closer the price will be to the underlying. So if the underlying stays exactly the same, each point on the curve in Figure 8.1 would have to slowly move down until it finally hits the underlying asset's price.

The reverse is therefore true for backwardation, which then has a built-in bullish bias. The underlying would need to move down simply for the points on the backwardation structure to stay the same. If the underlying does not move, the contracts need to come up to meet it.

The chart shown in Figure 8.3 shows a theoretical situation, where a futures contract starts out in a state of contango. As you see, the price would then start above that of the underlying market, trading at a premium. As you see here, if nothing else changes, then for each day the contango contract has to move a tiny bit down, getting a tiny bit closer to the underlying. We already know that the two will by necessity be the same at the date of the expiry, so it will slowly approach that point.

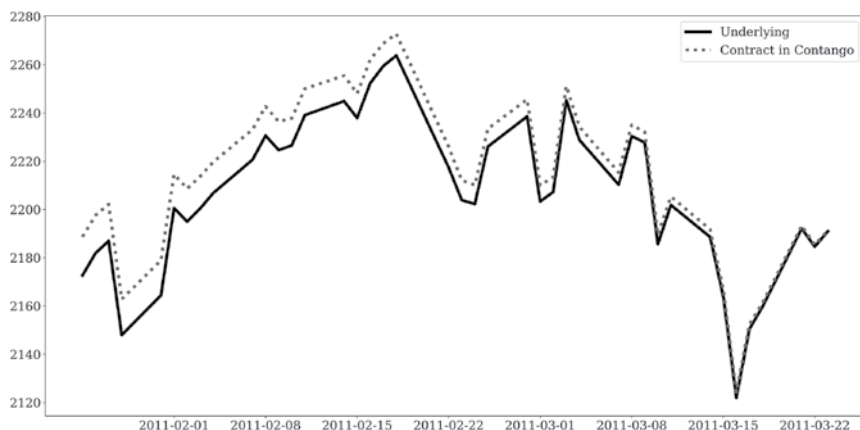


FIGURE 8.3 Contract in contango.